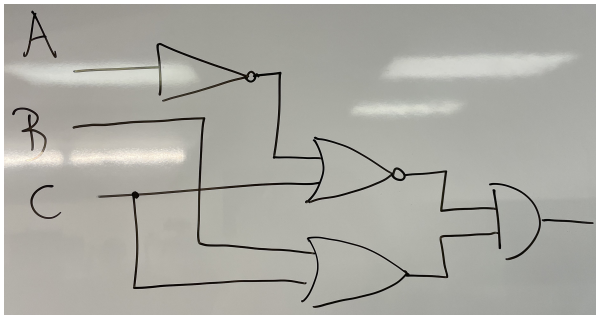
Drew University - CSCI 260 - Fall 2024
In-Class Exercise 2 - Digital Logic

Instruction: Please work in groups of 4 and write down your names for attendance. 09/20/2024
Names:

1. For the logic expression NOT(C OR NOT(A)) AND (C OR B), first give its equivalent logic circuit and then use a truth table to evaluate it.

A	B	C	NOT(A)	$x = C$ OR NOT(A)	NOT(x)	C OR B	NOT(x) AND (C OR B)
0	0	0	1	1	0	0	0
0	0	1	1	1	0	1	0
0	1	0	1	1	0	1	0
0	1	1	1	1	0	1	0
1	0	0	0	0	1	0	0
1	0	1	0	1	0	1	0
1	1	0	0	0	1	1	1
1	1	1	0	1	0	1	0

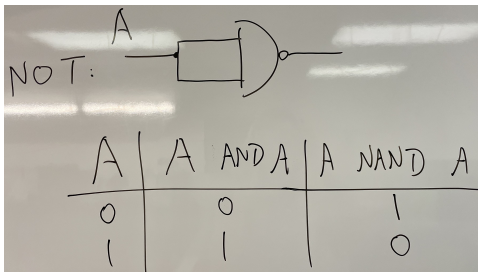


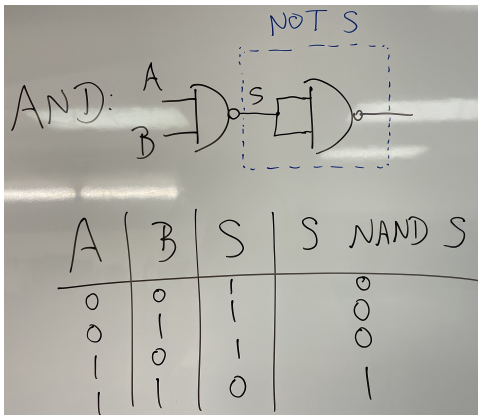
2. Show that the NAND gate, by itself, is logically complete. To do so, construct logical circuits (i.e., diagrams) that perform the NOT, AND, and OR operations by only using the NAND gate.

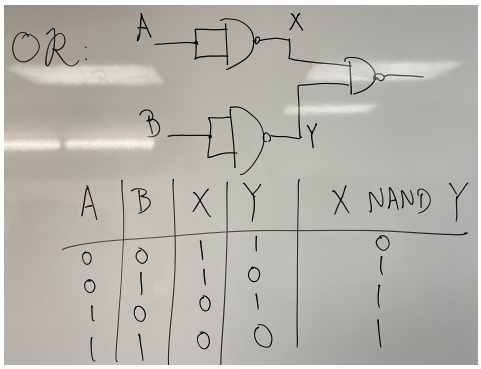
   Ans:

   - To construct NOT(A), we perform A NAND A.
   - To construct A AND B, we first let $S = A$ NAND B, and then perform S NAND S.
   - To construct A OR B, since we already constructed NOT and AND, we apply De Morgan's Law; i.e., NOT(NOT(A) AND NOT(B)).

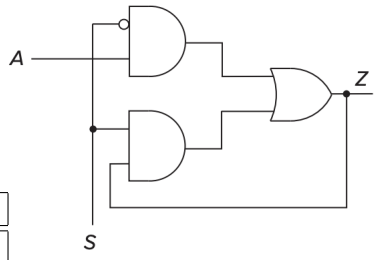
   You can draw the diagrams for these circuit; one thing to notice when you are drawing is that, once done with the diagrams for NOT and AND, we can abstract them as blackboxes.







3. For the following logic circuit below, answer the following questions.



(a) Describe its output Z using a truth table when the select signal S is a logical 0.

Ans: Let the result of the top AND gate be X and the result of the bottom AND gate be Y.

A	S	$\neg S$	X	Y	Z
0	0	1	0 [A is 0]	0 [S is 0]	0 [X, Y are 0]
1	0	1	1 [$A, \neg S$ are 1]	0 [S is 0]	1 [X is 1]

The take away is that, $Z = A$.

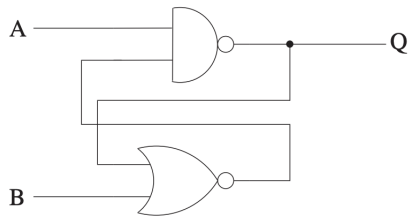
(b) If the select signal S is switched from a 0 to 1, what will the output be?

Ans: Denote Z as the current output value holds and Z' the one it changes to after S changes.

A	S	$\neg S$	X	Y	Z	Z'
0	1	0	0	same as Z	0	0
					1	1
1	1	0	0	same as Z	0	0
					1	1

The take away is that, Z stays the same.

4. Given the latch below, for what values of A and B will the latch remain in the quiescent/quiet state (i.e., the output will not change)?



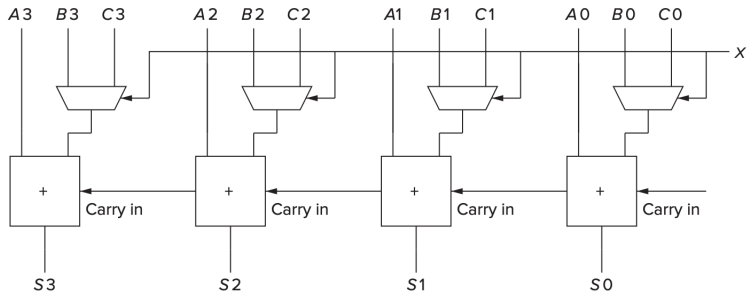
Ans: Let the result of the top NAND gate be X and the result of the bottom NOR gate be Y.

Note that, for $A = 1, B = 0$ case, need to discuss different Q cases to work back towards X and Y.

A	B	X	Y	Q
0	0	1	0	1
0	1	1	0	1
1	0	0	1	0
		1	0	1
1	1	1	0	1

All the combination of A, B give quiet state.

5. The figure below shows a logic circuit that appears in many of today's processors. Each of the boxes is a full-adder circuit and each of the trapezoids is a 2-to-1 MUX. What does the value on the wire X do?



Ans: X is used for selecting the second operand between $B = B_3B_2B_1B_0$ and $C = C_3C_2C_1C_0$ that is to be added to $A = A_3A_2A_1A_0$.

6. Implement a 4-to-1 MUX using only 2-to-1 MUXes. You can treat the 2-to-1 MUX as a blackbox (i.e., no need to show the gates), but when implementing, make sure you connect all the necessary wires. Also, write out the truth table of the 4-to-1 MUX for selecting input. **Show all your work**.

   Remember that a 4-to-1 MUX has 4 input sources, 2 control signals, and 1 output. (Hint: How do we make a 3-input AND gate out of 2-input AND gates?)

   Ans: Define 3 MUXes, say, P, Q, R.

   Let the input of P be A, B and control signal S_0; the input of Q be C, D and control signal S_0. Let the output of be P, Q be X, Y, respectively.

   Then, let the input of R be X, Y and control signal S_1; and its output be Z.

   Say for each control signal, being 0 is the same as selecting the first input, and being 1 is the same as selecting the second input. The truth table is then:

S_0	S_1	X	Y	Z
0	0	A	C	A
0	1	A	C	C
1	0	B	D	B
1	1	B	D	D

